

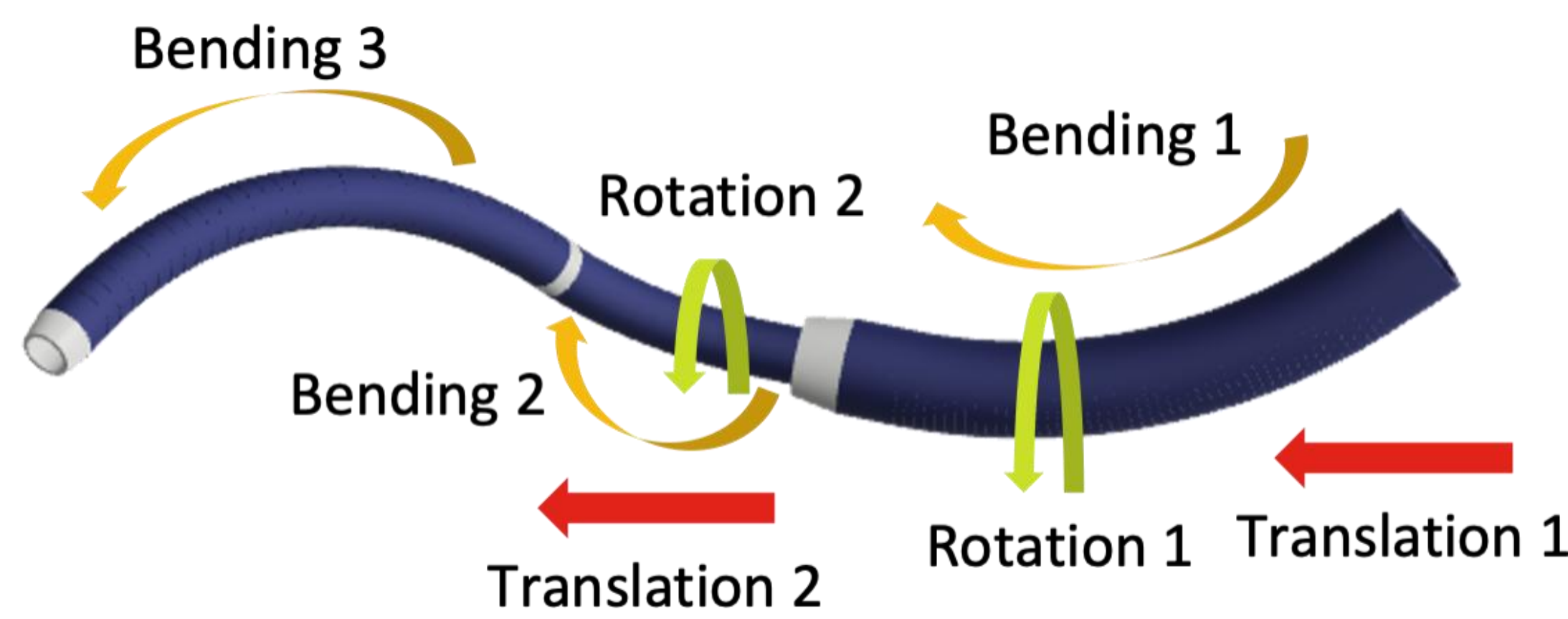
From Perception to Action: Data-Driven Catheter Shape Control for Robotic Cardiovascular Interventions

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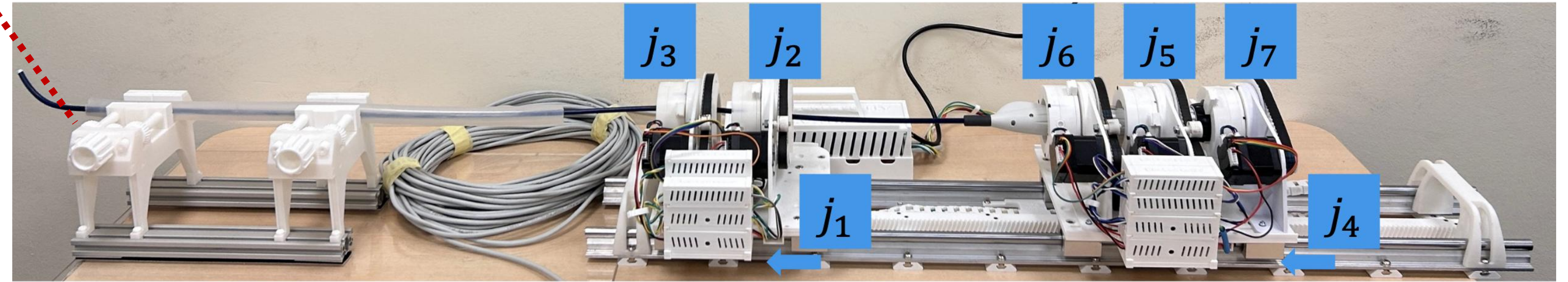
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The corresponding movement of the catheter tip in response to handle manipulations.



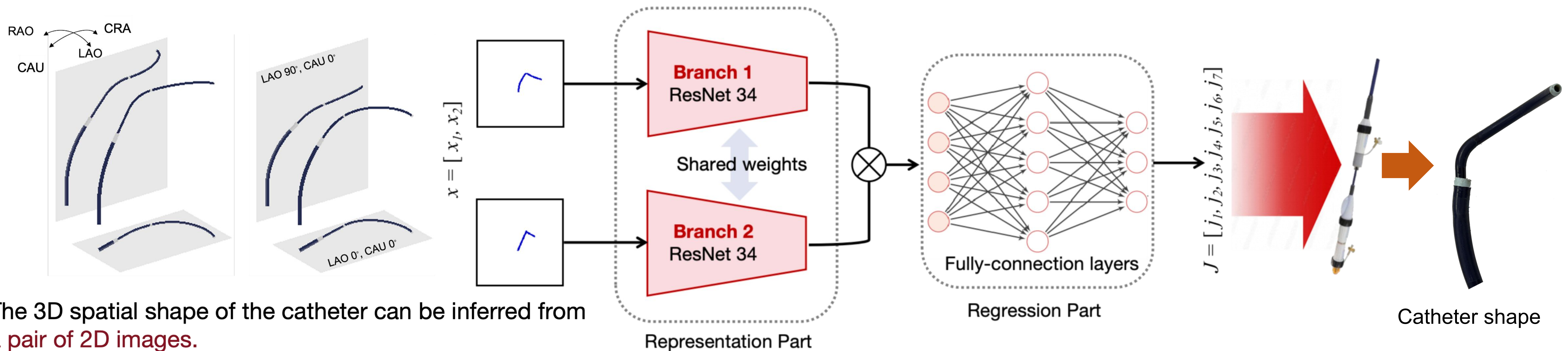
7-DOF Robotic System



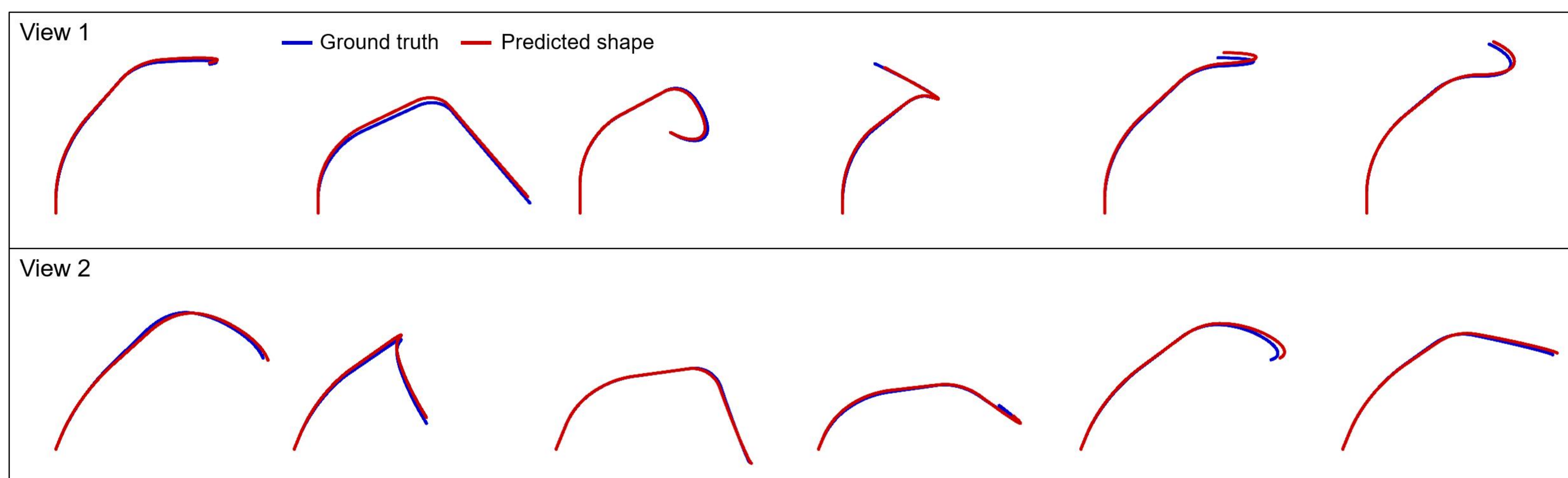
Motivation: Existing interventional robots feature cumbersome operation and poor repeatability. We construct a data-driven model based on catheter digital twins to realize direct mapping from images to robot actuation parameters.

Method and Framework

We split 3D catheter deformation into two 2D representations. With MATLAB simulations, we created a public dataset of 89,000 paired samples and trained a Siamese CNN to realize mapping from catheter shapes to robotic movements.

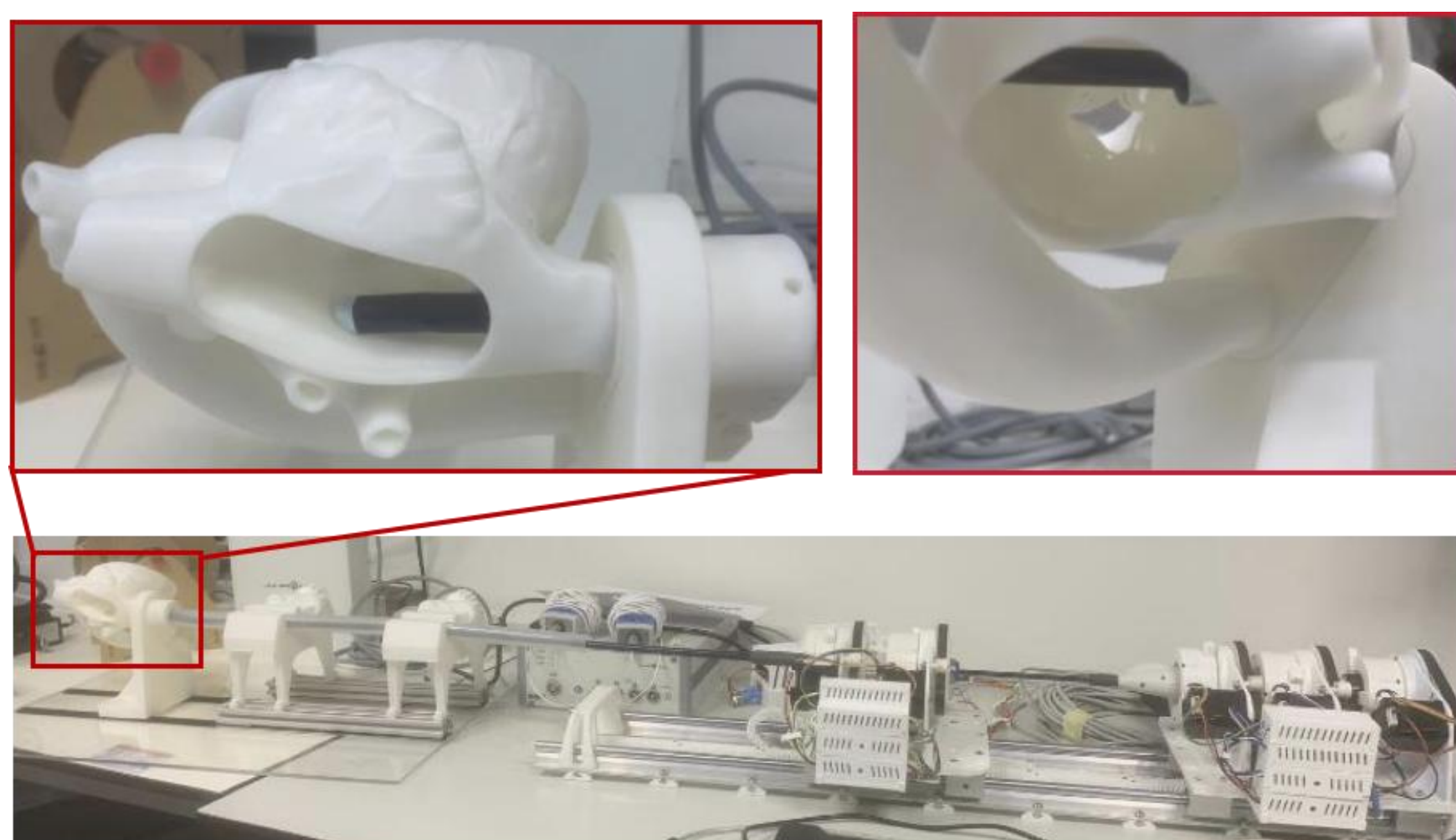


Experimental Results



Our method achieves high-fidelity control (relative errors from 1.846% to 5.402%), and provides novel insights for perception-to-action intervention robotics.

Joint	Avg. Err.	Rel. Err.	Var.
Joint 1 (Translation 1)	1.046 mm	5.230%	0.965
Joint 2 (Rotation 1)	1.661°	1.846%	2.477
Joint 3 (Bending 1)	1.299°	3.023%	1.049
Joint 4 (Translation 2)	0.795 mm	3.975%	0.413
Joint 5 (Rotation 2)	1.919°	3.198%	2.779
Joint 6 (Bending 2)	2.027°	3.534%	2.807
Joint 7 (Bending 3)	6.331°	5.402%	26.723



Shape comparison in free space



Conclusion and Future Works

We build a catheter digital twin driven Siamese CNN for direct image-based robotic catheter control and release an open 89,000-sample paired dataset to support the community. We will build real-time intraoperative digital twin, add biomechanical constraints, and validate the system via in-vivo animal.

